CSC 4740/6740 **Data mining** **D**ue by: Sept 27.

Name:

Assignment 2

Q1: Explain the following *normalization methods by using formula,* (5 points)

(a) min-max normalization to [0,1]

(b) z-score normalization/Standardization

(c) apply normalization to the data vector [200, 300, 400, 600, 1000]

(c1) min-max normalization by setting *min* = 0 and *max* = 1

(c2) z-score normalization

(d) now a new data point was collected value =250, Please use the **SAME Z transformation** to transform the new data.

Q2: The following data present movie genre liked by numbers of participants from UK and USA. (values are made up) / Can you answer whether the country is related or independent with preference of movie genre using Chi-square test, p value of 0.05? (**10 point**)

	Comedy	Action	Suspension	Romance	Science fiction
UK	120	300	250	230	100
USA	250	350	260	290	350

Chi-Square Table

TABLE 24.1 To be significant at level α, a chi-square statistic must be larger than the table entry for α

	Significance Level α						
df	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.01**	**0.001**
1	1.32	1.64	2.07	2.71	3.84	6.63	10.83
2	2.77	3.22	3.79	4.61	5.99	9.21	13.82
3	4.11	4.64	5.32	6.25	7.81	11.34	16.27
4	5.39	5.99	6.74	7.78	9.49	13.28	18.47
5	6.63	7.29	8.12	9.24	11.07	15.09	20.51
6	7.84	8.56	9.45	10.64	12.59	16.81	22.46
7	9.04	9.80	10.75	12.02	14.07	18.48	24.32
8	10.22	11.03	12.03	13.36	15.51	20.09	26.12
9	11.39	12.24	13.29	14.68	16.92	21.67	27.88

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Q3: Warm-up with PCA on the Palmer Penguins Dataset (20 pts)

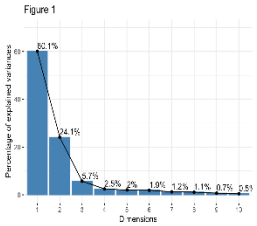
Dataset: Palmer Penguins — 344 penguins (Adelie, Chinstrap, Gentoo) measured on 3 islands in the Palmer Archipelago, Antarctica. Features: bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g, plus species, island, sex.

If you want to know more about the data source https://allisonhorst.github.io/palmerpenguins/

Download: penguins.csv (raw CSV, GitHub)

Tasks

1. **Load and inspect.** Load the dataset and report: number of rows, number of missing values per column, and the count of penguins per species.
2. **Handle missing data.**
   - Choose a strategy and apply to the data : drop rows with any missing numeric feature (dropna), or impute (e.g., mean/median imputation, possibly per-species).
   - **Written answer:** Which approach did you choose, and why? Discuss at least one thing you'd lose or risk with each of the following: (a) dropping rows, (b) imputing with the global mean, (c) imputing with the per-species mean. Hint: Consider how many rows are actually affected — does the scale of the missingness change which option is reasonable?
3. **Standardize the features.**
   - Standardize the four numeric features (zero mean, unit variance) before running PCA.
   - **Written answer:** Why does standardization matter here specifically? (Hint: look at the raw scales and variances of body_mass_g vs. bill_depth_mm before scaling) Is there a general rule for when standardization is necessary vs. optional before PCA?
4. Compute PCA on the standardized features.
   - **Plot the scree plot with all possible PCs** (explained variance ratio per component, and cumulative). Like



   - **Cross-correlation matrices — before and after PCA.** Compute and plot (e.g., as a heatmap) the correlation matrix of the four **standardized input features** (bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g). Then compute the correlation matrix of the **principal component scores** (all 4 PCs). Plot both heatmaps side by side.
   - **Written answer:** What do you notice about the off-diagonal entries of the PC correlation matrix compared to the input feature correlation matrix, and why does this happen? Which pairs of *original* features were most strongly correlated?
5. **Written answer:** How many components are needed to explain 90% of the variance? Rerun PCA on the *unstandardized* features and compare — does the answer change? Why?

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6. Plot the 2D PCA projection (PC1 vs. PC2), colored by species. Comment on which species separate cleanly and which overlap, and connect this to the PCA loadings (which original features dominate PC1 vs. PC2?).

   Written answer: Based on the loadings, give a one-sentence physical interpretation of what PC1 represents and what PC2 represents (e.g., is PC1 mostly capturing overall body size? Is PC2 capturing something about bill shape independent of size?). Does this interpretation match which species PC1 vs. PC2 separates well?

Q4: Scaling Up: PCA on High-Dimensional Data (20 pts)

Dataset: Digits — 1,797 images of handwritten digits (0–9), each 8×8 pixels (64 features), a small/clean subset of the UCI "Optical Recognition of Handwritten Digits" dataset.

Download/Import data: Digits:

```
from sklearn.datasets import load_digits

digits = load_digits()

X, y = digits.data, digits.target   # X: (1797, 64), y: (1797,) digit labels 0-9
```

If you'd rather work with the raw source files directly (e.g., to see the original format), the original dataset is hosted at UCI: Optical Recognition of Handwritten Digits Data Set (see optdigits.tra / optdigits.tes for the raw train/test files).

Tasks

1. Load Digits. Report the shape of the data, the number of classes, and reshape a few samples back to 8×8 to visualize them as images (plt.imshow) to confirm you're looking at real digits.
2. Run PCA to 50 components. Plot scree plot and cumulative explained variance.
3. Visualize the first 2 PCA components colored by label. Discuss what's lost compared to the full 50-dimensional representation.
4. Reconstruct 5 random sample images from only the top *k* PCs for *k* = 5, 20, 40, Show the reconstructions side-by-side with the originals and discuss the quality/dimension trade-off.

Q5: t-SNE (15 pts)

1. Run t-SNE (via sklearn.manifold.TSNE or openTSNE for speed) on the **raw high-dim data** and separately on the **PCA-reduced (50-dim) data,** to reduce to 2 dimension data with perplexity of 5. Compare runtime and resulting embeddings. (If your computer has limited power, then **just using PCA-reduced data**.)
2. Sweep perplexity over at least 4 values (e.g., 5, 30, 50, 100). Plot the resulting embeddings in a grid and discuss how cluster shape, size, and separation change.
3. Run t-SNE twice with the same perplexity but different random seeds. Compare the two embeddings. What does this tell you about interpreting **distances between clusters** and **cluster sizes** in t-SNE plots?

Q6 : UMAP (15 pts)

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1. Run UMAP (umap-learn) on the same PCA-reduced data used for t-SNE to reduce into 2 dimension, with different n_neighbors (e.g., 5, 15, 50, 200) and min_dist (e.g., 0.0, 0.1, 0.5, 0.9) as a 2D grid (at least 4 combinations shown). Discuss the effect of each parameter separately.

Q7: Written answer: Answer in ~300 words total: **(15 pts)**

1. If you had to explain to a biologist collaborator why their tSNE/UMAP plot's cluster sizes and inter-cluster distances shouldn't be over-interpreted, what would you say?
2. Give one concrete scenario where PCA would be the *better* choice over UMAP/t-SNE, and one where UMAP/t-SNE would be clearly better than PCA.
2. Both t-SNE and UMAP are stochastic and sensitive to hyperparameters. What practical steps would you take in a real project to make sure your conclusions aren't an artifact of a lucky/unlucky random seed or parameter choice?